

Topic

- Why K-Means & Hierarchical fail
 - DBSCAN concept
 - Density-based clustering
 - eps & min_samples
 - Outlier detection
 - Practical implementation
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Class Objective

Students will:

- Understand limitations of K-Means
 - Learn **DBSCAN (Density-Based Clustering)**
 - Understand core concepts:
 - eps
 - min_samples
 - Detect **outliers/noise**
 - Apply DBSCAN practically
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1. CLASS START (Hook)

Start like this:

“K-Means assumes clusters are circular...
But what if data shape is irregular?”

 Pause

Then:

“Today we learn DBSCAN — which can handle any shape and even detect outliers.”

2. PROBLEM WITH K-MEANS

Issues

-  Assumes circular clusters
 -  Must choose K
 -  Cannot detect outliers
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Teaching Line

“K-Means fails when clusters are irregular or noisy.”

3. 📖 WHAT IS DBSCAN?

Definition

DBSCAN is a clustering algorithm that groups data based on **density**.

Simple Meaning

“Dense areas form clusters, sparse areas are treated as noise.”

4. 📖 KEY CONCEPTS

◆ eps (epsilon)

👉 Radius around a point

- Defines neighborhood
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◆ min_samples

👉 Minimum number of points required to form a cluster

5. 📖 TYPES OF POINTS

◆ Core Point

- Has $\geq$ min_samples within eps
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◆ Border Point

- Near a cluster but not dense
-

◆ Noise (Outlier)

- Not part of any cluster
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💡 Teaching Line

“DBSCAN automatically identifies outliers.”

6. HOW DBSCAN WORKS

Step-by-Step

1. Pick a point
 2. Check neighbors within eps
 3. If dense → create cluster
 4. Expand cluster
 5. Mark sparse points as noise
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7. ADVANTAGES

- ✓ No need to choose K
 - ✓ Handles irregular shapes
 - ✓ Detects outliers
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8. LIMITATIONS

- ✗ Choosing eps is tricky
 - ✗ Struggles with varying densities
 - ✗ Sensitive to parameters
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9. K-MEANS vs DBSCAN

Feature	K-Means	DBSCAN
Shape	Circular	Any shape
Need K	Yes	No
Outliers	No	Yes
Concept	Distance	Density

10. INTERVIEW CONCEPTS

Q1: What is DBSCAN?

👉 Density-based clustering algorithm.

Q2: What is eps?

👉 Radius for neighborhood.

Q3: What is min_samples?

👉 Minimum points to form cluster.

Q4: Why DBSCAN better than K-Means?

👉 Handles irregular shapes & detects outliers.

11. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Apply DBSCAN clustering

12. 📊 OUTPUT INTERPRETATION

- Cluster labels:
 - 0,1 → clusters
 - -1 → outliers
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👉 Example:

- (100,10) → noise
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👉 Highlight:

“DBSCAN automatically detects unusual data points.”

“DBSCAN is powerful because it can find clusters of any shape and automatically detect outliers — something K-Means cannot do.